



CV2090

Probability & Statistics for Civil Engineers

(Jul – Nov 2026)

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L25
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Extreme Value Distributions **4.2.3**

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Forms of Extreme Value Distributions

Messages from the Last Class

- **Message 1: Every record is an unobserved extreme.**

- Systems need to be designed for extremes, NOT yet observed
 - Dams for 1000-year flood
 - Buildings for 2,475-year earthquake
 - NPPs for 10,000-year earthquake

- **Message 2: Designing/planning for the observed maxima won't cut it!**

- E.g., 1953 North Sea Floods in The Netherlands
- 2011 Fukushima Nuclear Incident

- **Message 3: Parent dist. (initial variate) $\Rightarrow$ Extreme dist.**

- Tail of parent dist. can lead to Asymptotic EVD
 - Exponentially decaying tail $\rightarrow$ Gumbel, Type I, Double Exponential
 - Polynomially decaying tail $\rightarrow$ Fréchet, Type II, Single Exponential
 - Bounded tail $\rightarrow$ Rev. Weibull, Exponential with a bound

Messages from the Last Class



Levee (man made)



Levee (Natural)

(Surge)
Storm Barrier
Rotterdam, ND



Fisher-Tippett-Gnedenko Theorem

• Extreme value theorem

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- Let $X_1, X_2, \dots, X_n$ be an n -sized sample of i.i.d.'s (independent and identically-distributed rv's), each with the CDF $F_X(x)$
- Suppose $a_n > 0$ and $b_n \in \mathbb{R}$ such that the following limits converge

$$\lim_{n \rightarrow \infty} \Pr \left(\frac{\max\{X_1, \dots, X_n\} - b_n}{a_n} \leq y \right) \equiv \lim_{n \rightarrow \infty} F_{Y_n}(y) = G_{\max}(y)$$

- Normalized maximum converges to the GEV (generalized ext. val.) CDF

$$G_{\max}(y) = \exp \left[- \left(1 + \xi \frac{y - u}{\alpha} \right)^{-1/\xi} \right]$$

Gumbel for maxima ($\xi = 0$): $G_{\max}(y) = \exp[-e^{-\alpha_n(y-u_n)}]$

Fréchet for maxima ($\xi > 0$): $G_{\max}(y) = \exp[-\{(y-u)/\alpha\}^{-k}]$ for $y > u$

Rev. Weibull ($\xi < 0$): $G_{\max}(y) = \exp[-\{(w-y)/\alpha\}^k]$ for $y \leq w$

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Fisher-Tippett-Gnedenko Theorem

$$\lim_{n \rightarrow \infty} \Pr \left(\frac{\min\{X_1, \dots, X_n\} - b_n}{a_n} \leq y \right) \equiv \lim_{n \rightarrow \infty} F_{Y_1}(y) = G_{\min}(y)$$

- Normalized minimum converges to the GEV (generalized ext. val.) CDF

$$G_{\min}(y) = 1 - \exp \left[- \left(1 + \xi \frac{y - u}{\alpha} \right)^{-1/\xi} \right]$$

Gumbel for Minima ($\xi = 0$): $G_{\min}(y) = 1 - \exp[-e^{\alpha_1(y-u_1)}]$

Fréchet for Minima ($\xi > 0$): $G_{\min}(y) = 1 - \exp \left[- \left(\frac{u-y}{\alpha} \right)^{-k} \right]$ for $y < u$

Standard Weibull ($\xi < 0$): $G_{\min}(y) = 1 - \exp \left[- \left(\frac{y-\varepsilon}{w-\varepsilon} \right)^k \right]$ for $y \geq \varepsilon$

Asymptotic EVD is determined solely by the tail of the parent CDF $F_X(x)$

Extreme Value Theorem

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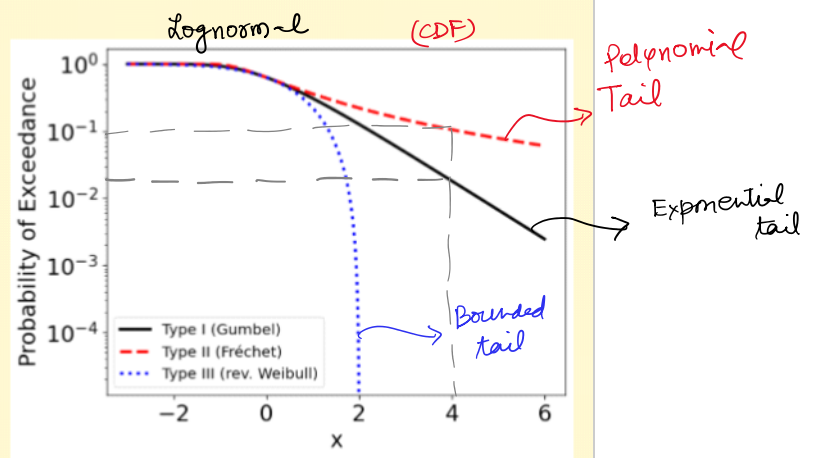
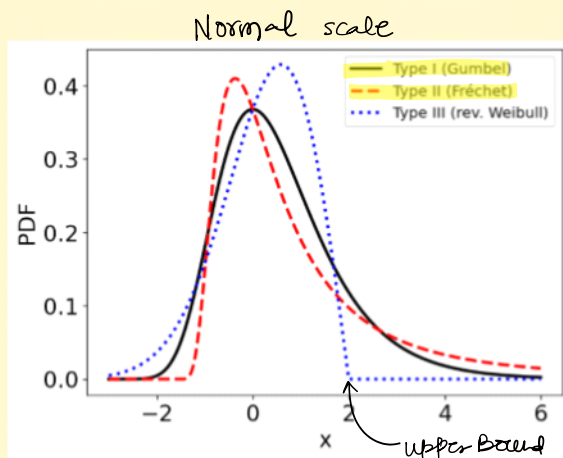
"Central Limit Theorem of Extremes"

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 $X \rightarrow \xi$
 $Z \rightarrow \zeta$
 $T \rightarrow \tau$

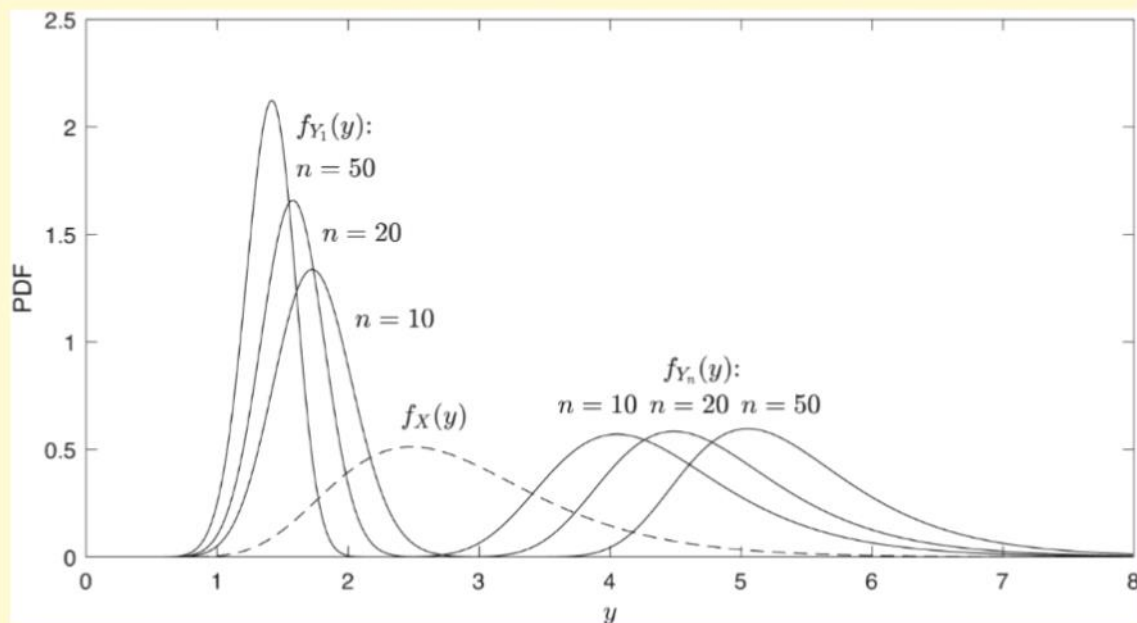
Tailedness decides the extreme family

Parent distribution	Tail type	Extreme value type
Normal (exponential tail); $\xi = 0$	Medium-/baseline tailed	Gumbel (double exp.)
t-distribution (poly tail); $\xi > 0$	Heavy-/fat-tailed	Fréchet (exp.)
Uniform/beta (bounded tail); $\xi < 0$	Light-/thin-tailed	Rev. Weibull (U-bound exp.) Std. Weibull (L-bound exp.)



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Gumbel distributions



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Extreme value analysis

- Magnitude of extremes

Generalized Extreme Value (GEV)

- Frequency of extremes

Poisson "process"

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Return period

- Four key concepts

prob. of exceedance in 1 year: $P_{f,1y}$
Return Period: $T = 1/P_{f,1y}$
Design Life: T_d , lifetime of a structure
prob of exceedance in T_d : P_{f,T_d}

- Poisson distribution

prob. of observing x number of events in T years

$$Pr(x; \lambda, T) = \frac{(\lambda T)^x e^{-\lambda T}}{x!}$$

$$E[X] = \lambda T = \text{Var}[X]$$

$$Pr(x=0) = e^{-\lambda T}$$

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Return period

- A design event has **10% probability of exceedance in 50^T years** ($p_{f,50y} = 0.10$). What is the **annual expected number of events λ** and the **return period T_r** in years?

$$\Pr(x, \lambda, T) = \frac{(\lambda T)^x e^{-\lambda T}}{x!}$$

$$\Pr(\text{exceedance}) = 1 - \Pr(\text{Non-exceedance})$$

$$\Rightarrow p = 1 - \Pr(\text{Non-exceedance}) = 1 - e^{-\lambda T}$$

$$\Rightarrow \Pr(\text{Non-exceedance}) = 1 - p$$

$$\Rightarrow \lambda = -\frac{\ln(1-p)}{T}$$

$$\left. \begin{array}{l} p = 10\%, T = 50y \\ \lambda = 2.107 \times 10^{-3}, 1/\lambda = 475 \text{ yr} \end{array} \right\} \begin{array}{l} \lambda = 4.04 \times 10^{-4}, T_r = 2475 \text{ yr} \end{array}$$

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Return period

- Given a design life of $T_d = 50$ years and a design event with annual probability ($p_{f,1y} = 1/475$), what is the **probability of failure during the design life, $p_{f,Td}$** ?

$$\Pr(x, \lambda, T) = \frac{(\lambda T)^x e^{-\lambda T}}{x!}$$

$$\Pr(\text{failure in } T_d) = 1 - \Pr(\text{Non-failure in } T_d)$$

$$\Rightarrow p_{f,Td} = 1 - e^{-\lambda T_d}$$

$$\lambda = 1/475, T_d = 50 \text{ y}$$

$$p_{f,Td} = 10\%$$

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Return period

- What is the probability of a 475-year return period event occurring at least once in 475 years?

$$\Pr(x, \lambda, T) = \frac{(\lambda T)^x e^{-\lambda T}}{x!}$$

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Thank you!